

Periodic table

The **periodic table**, also known as the **periodic table of the elements**, is an ordered arrangement of the **chemical elements** into rows ("periods") and columns ("groups"). An icon of **chemistry**, the periodic table is widely used in **physics** and other sciences. It is a depiction of the **periodic law**, which states that when the elements are arranged in order of their **atomic numbers** an approximate **recurrence** of their properties is evident. The table is divided into four roughly rectangular areas called **blocks**. Elements in the same group tend to show similar chemical characteristics.

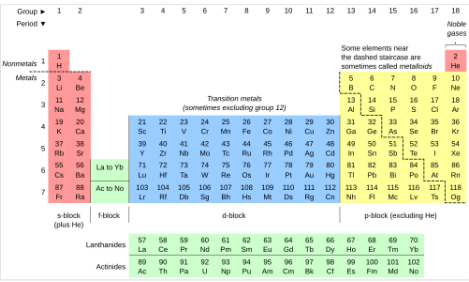
Vertical, horizontal and diagonal **trends** characterize the periodic table. **Metallic** character increases going down a group and from right to left across a period. **Nonmetallic** character increases going from the bottom left of the periodic table to the top right.

The first periodic table to become generally accepted was that of the Russian chemist **Dmitri Mendeleev** in 1869; he formulated the periodic law as a dependence of chemical properties on atomic mass. As not all elements were then known, there were gaps in his periodic table, and Mendeleev successfully used the periodic law to **predict some properties of some of the missing elements**. The periodic law was recognized as a fundamental discovery in the late 19th century. It was explained early in the 20th century, with the discovery of atomic numbers and associated pioneering work in quantum mechanics, both ideas serving to illuminate the internal structure of the **atom**. A recognisably modern form of the table was reached in 1945 with **Glenn T. Seaborg's** discovery that the **actinides** were in fact f-block rather than d-block elements. The periodic table and law have become a central and indispensable part of modern chemistry.

The periodic table continues to evolve with the progress of science. In nature, only elements up to atomic number 94 exist;^[a] elements beyond that can only be **synthesized** in the laboratory. By 2010, the first 118 elements were known, thereby completing the first seven rows of the table;^[1] however, chemical characterization is still needed for the heaviest elements to confirm that their properties match their positions. New discoveries will extend the table **beyond these seven rows**, though it is not yet known how many more elements are possible; moreover, theoretical calculations suggest that this unknown region will not follow the patterns of the known part of the table. Some scientific discussion also continues regarding whether some elements are correctly positioned in the table. Many **alternative representations** of the periodic law exist, and there is some discussion as to whether there is an optimal form of the periodic table.

Structure

Periodic table																		
Group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 Period	Hydrogen & alkali metals	Alkaline earth metals											Triels	Tetrels	Pnictogens	Chalcogens	Halogens	Noble gases
	Hydrogen																	
	1 H 1.0080																	
	Lithium	Beryllium											Boron	Carbon	Nitrogen	Oxygen	Fluorine	
	3 Li 6.94	4 Be 9.0122											5 B 10.81	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	
	Sodium	Magnesium											Aluminium	Silicon	Phosphorus	Sulfur	Chlorine	
	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.085	15 P 30.974	16 S 32.06	17 Cl 35.45	
2	Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc	Gallium	Germanium	Arsenic	Selenium	Bromine	
3	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.38	31 Ga 69.723	32 Ge 72.630	33 As 74.922	34 Se 78.971	35 Br 79.904	
4	Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium	Indium	Tin	Antimony	Tellurium	Iodine	
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.95	43 Tc [97]	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.905	
6	Caesium	Barium	* Lutetium	* Hafnium	* Tantalum	* Tungsten	* Rhenium	* Osmium	* Iridium	* Platinum	* Gold	* Mercury	* Thallium	* Lead	* Bismuth	* Polonium	* Astatine	
7	55 Cs 132.91	56 Ba 137.33	* 71 Lu 174.97	* 72 Hf 178.49	* 73 Ta 180.95	* 74 W 183.84	* 75 Re 186.21	* 76 Os 190.23	* 77 Ir 192.22	* 78 Pt 195.08	* 79 Au 196.97	* 80 Hg 200.59	* 81 Tl 204.38	* 82 Pb 207.2	* 83 Bi 208.98	* 84 Po [209]	* 85 At [210]	
	Francium	Radium	* Lawrencium	* Rutherfordium	* Dubnium	* Seaborgium	* Bohrium	* Hassium	* Meitnerium	* Darmstadtium	* Roentgenium	* Copernicium	* Nihonium	* Flerovium	* Moscovium	* Livermorium	* Tennessine	
	87 Fr	88 Ra	* 103 Lr	* 104 Rf	* 105 Db	* 106 Sg	* 107 Bh	* 108 Hs	* 109 Mt	* 110 Ds	* 111 Rg	* 112 Cn	* 113 Nh	* 114 Fl	* 115 Mc	* 116 Lv	* 117 Ts	



Periodic table of the chemical elements showing the most or more commonly named sets of elements (in periodic tables), and a traditional dividing line between metals and nonmetals. The f-block actually fits between groups 2 and 3; it is usually shown at the foot of the table to save horizontal space.